EduPro Research Paper

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1. ## Introduction and Objectives;

Introduction

EduPro is an online education platform that needs a structured, data-driven approach to evaluate instructor performance and course quality. Without this approach, it is difficult to identify highly rated instructors, understand differences in course ratings, and determine where improvements are needed. This study analyses instructor profiles, course ratings, and recorded enrollment to support evidence-based decisions.

Objectives

- Examine the distribution of instructor ratings and identify high-rated and low-rated instructors.
- Explore relationships between teaching experience, instructor ratings, and linked course ratings.
- Compare course quality across categories, levels, and instructor expertise areas.
- Compare recorded enrollment volumes across instructor rating tiers.
- Examine variation in linked course ratings and recommend improvements

2. ## Dataset and Data Validation:

Sheet	Records	Information used
Teachers	60	Instructor ID, name, age, gender, expertise, experience, and rating
Courses	60	Course ID, name, category, level, and rating
Transactions	10,000	Transaction ID, CourseID, and TeacherID

Teachers and Transactions were connected using **TeacherID**, while Courses and Transactions were connected using **CourseID**. The integrated table, **Qanalysis**, contains one row per transaction. Removing repeated TeacherID–CourseID combinations produced **887 unique instructor-course pairs** for analysing course relationships.

Data Validation

Validation checks found no missing linking IDs, duplicate TeacherIDs or CourseIDs in the source dimension sheets, or transactions with unmatched instructor or course IDs. The integrated table retained all 10,000 transactions.

All 60 courses were linked to multiple instructors, ranging from **7 to 30 instructors per course**. Consequently, a course's overall rating cannot be attributed to one instructor. Transactions are treated as recorded enrollments and do not establish attendance or course completion.

3. Methodology:

The study used exploratory data analysis (EDA) to describe instructor performance, course quality, and recorded

enrollment patterns. Excel Power Query was used to join the Teachers, Courses, and Transactions sheets. PivotTables, formulas, charts, and conditional formatting were used to summarise and visualise the results.

Three units of analysis were used:

- **Individual instructors: Each of the 60 instructors was counted once when analysing profiles, teacher ratings, and experience versus teacher rating.**
- **Unique courses: Each of the 60 courses was counted once when comparing ratings across course categories and levels.**
- **Instructor-course pairs: The 887 unique pairs were used to examine linked course ratings by instructor, expertise, rating tier, and gender versus course level. Transactions were used to count recorded enrollments.**

Pearson correlation measured the direction and strength of linear relationships between experience and teacher rating, experience and linked course rating, and teacher rating and linked course rating. These correlations describe associations and do not establish causation.

Instructors were grouped into three rating tiers: Low—below 3; Mid—3 to below 4; High—4 and above. Average linked course ratings and enrollment volumes were compared across these tiers. Enrollment concentration was also examined to assess the influence of the two instructors with the most enrollments.

Population standard deviation measured variation in each instructor's linked course ratings. Lower standard deviation indicates less variation, but does not necessarily indicate higher quality. Because course ratings are shared across instructors, this measure describes course portfolios rather than personal teaching reliability.

The project's KPIs were defined as follows:

KPI	Definition
Average Teacher Rating	Mean rating across 60 instructors
Average Course Rating	Mean rating across 60 unique courses
Experience Impact Score	Pearson correlation between experience and teacher rating
Enrollment Influence Ratio	High-tier average enrollments per instructor divided by the platform average
Rating Consistency Index (proxy)	Mean of the 60 instructors' linked-course rating standard deviations

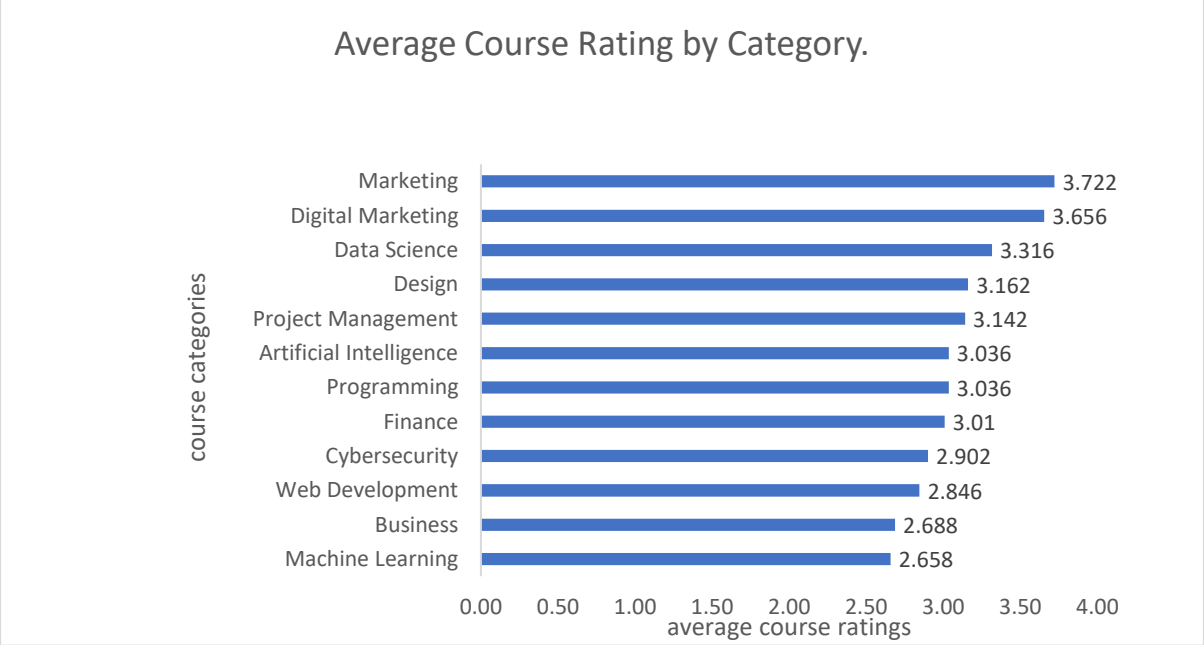
4.Analysis and Findings;

4.1 Instructor Ratings and Profiles

The average teacher rating was **3.13 out of 5**, with ratings ranging from **1.05 to 4.97**. Of the 60 instructors, **21 (35%)** belonged to the Low tier, **28 (46.67%)** to the Mid tier, and **11 (18.33%)** to the High tier. The Mid tier was the largest group, although it represented less than half of all instructors.

Yolanda Levine had the highest teacher rating (**4.97**), followed by **Kimberly Miller (4.58)**. **John Williams** had the lowest rating (**1.05**). These ratings identify instructors for recognition or further review, but do not explain the reasons behind their performance.

Most instructors were aged **30–49 years**, accounting for **80%** of instructors. Additionally, **51 instructors (85%)** had between **1 and 10 years of experience**. These figures describe the instructor population; age alone does not establish teaching quality.



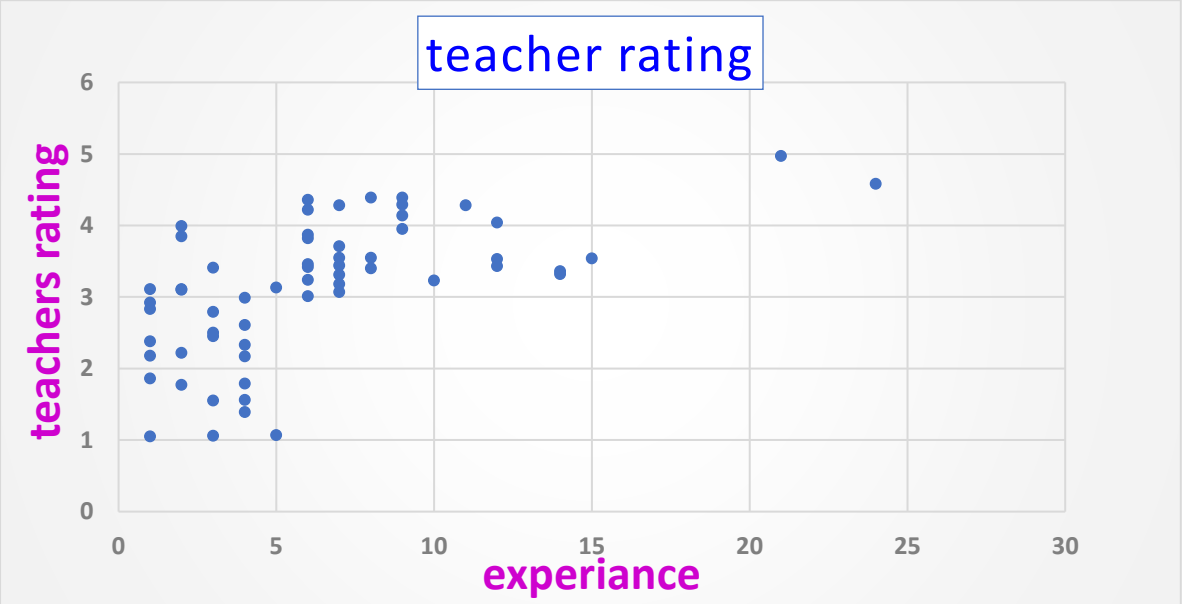
4.2 Experience and Ratings

Years of experience and teacher rating showed a positive Pearson correlation of **0.598** across 60 instructors. More experienced instructors tended to receive higher teacher ratings, although this relationship does not prove causation.

Average teacher ratings were **2.40** for the 1–5-year group, **3.71** for 6–10 years, **3.64** for 11–15 years, and **4.78** for 21–25 years. The pattern was not continuously increasing. The highest-experience group contained only **two instructors**, and no instructors fell in the 16–20-year group. Therefore, the data does not establish a reliable experience threshold or diminishing-return pattern.

Across the 887 instructor-course pairs, experience and linked course rating had a correlation of approximately **–0.013**, while teacher rating

and linked course rating had a correlation of approximately **−0.034**. Both indicate almost no linear relationship in this dataset. Shared course ratings limit conclusions about individual instructor impact.



4.3 Course Quality by Category and Level

Across 60 unique courses, the average course rating was **3.10 out of 5**. Marketing had the highest category average (**3.72**), followed by Digital Marketing (**3.66**). Machine Learning had the lowest average (**2.66**), followed by Business (**2.69**). Each category contained five courses. These averages identify areas for review, but do not establish the reasons for rating differences.

Intermediate courses averaged **3.32**, Beginner courses **3.19**, and Advanced courses **2.81**. The category-level heatmap revealed further differences. For example, Cybersecurity–Beginner scored **4.94**, while Machine Learning–Beginner scored **1.13**. Each of these combinations contained only one course, so neither result establishes consistent performance across multiple courses.

Average of CourseRating	course level

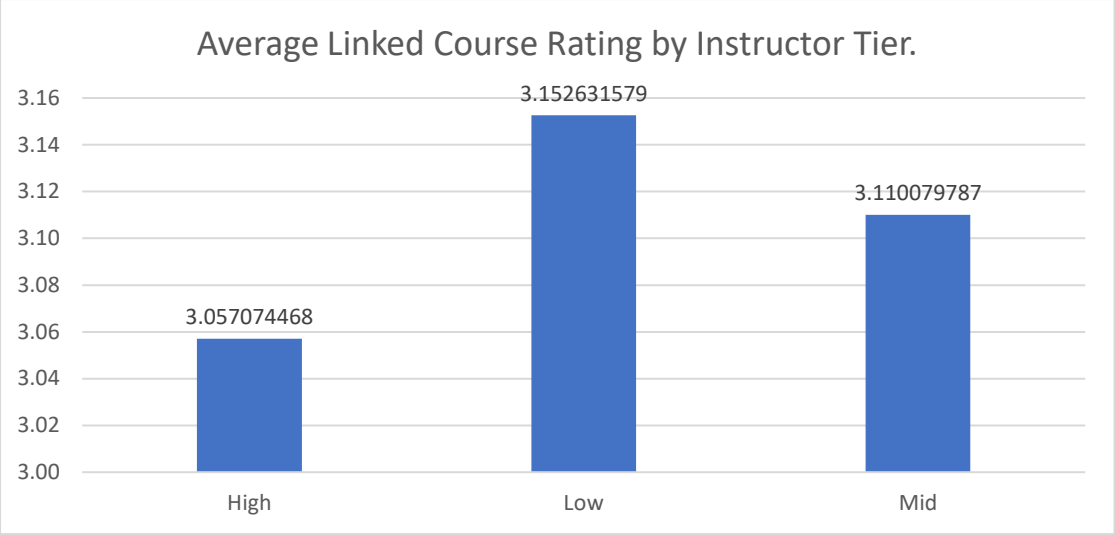
Row Labels	Advanced	Beginner	Intermediate
Artificial Intelligence	-	3.13	2.90
Business	2.93	1.96	2.69
Cybersecurity	2.38	4.94	2.40
Data Science	2.57	3.70	3.31
Design	2.83	4.51	-
Digital Marketing	3.52	3.29	4.66
Finance	1.59	3.31	3.43
Machine Learning	2.85	1.13	3.23
Marketing	3.26	-	4.42
Programming	-	3.19	2.43
Project Management	1.23	3.10	4.15
Web Development	2.90	2.76	-

4.4 Instructor Rating Tiers and Enrollments

Average linked course ratings were similar across instructor tiers: **High 3.06, Mid 3.11, and Low 3.15**. The difference between the highest and lowest averages was only **0.09 points**. Highly rated instructors were therefore not associated with higher average linked course ratings in this comparison.

High-tier instructors accounted for **6,820 enrollments (68.2%)**, averaging **620 per instructor**. However, Yolanda Levine and Kimberly Miller together accounted for **6,086 enrollments (60.86% of the platform total)**. Excluding these two, the remaining nine High-tier instructors averaged **81.56 enrollments**, compared with **76.38** for Low-tier instructors and **56.29** for Mid-tier instructors.

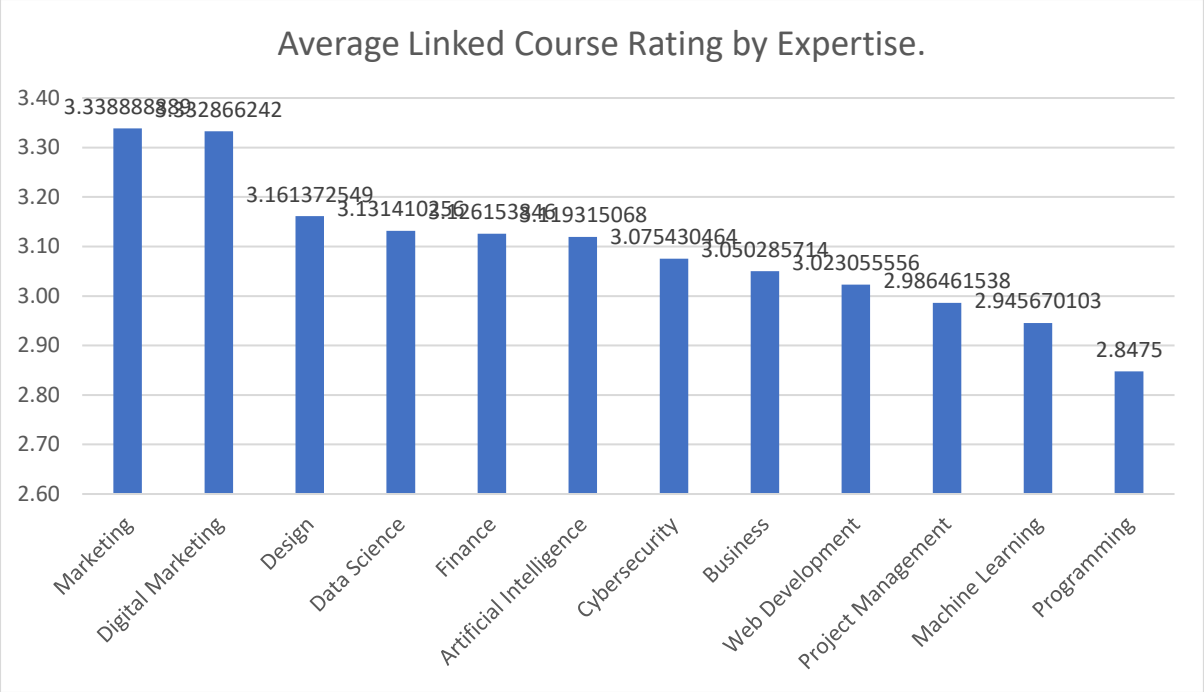
The Enrollment Influence Ratio was **3.72**, but the concentration among two instructors strongly influenced this result. The analysis does not establish that higher instructor ratings caused higher demand.



4.5 Expertise and Linked Course Quality

Marketing had the highest average linked course rating (**3.34 across 9 instructor-course pairs**), closely followed by Digital Marketing (**3.33 across 157 pairs**). Programming had the lowest average (**2.85 across 8 pairs**). Marketing's lead over Digital Marketing was only **0.01 points**, so it should not be interpreted as clearly superior performance.

These groups represent instructor expertise, which may differ from the categories of their linked courses. Pair counts are not learner review counts, and shared course ratings limit conclusions about teaching effectiveness. This comparison alone cannot establish which domains depend most on instructor quality or where instructor training is required.



4.6 Gender and Course Level

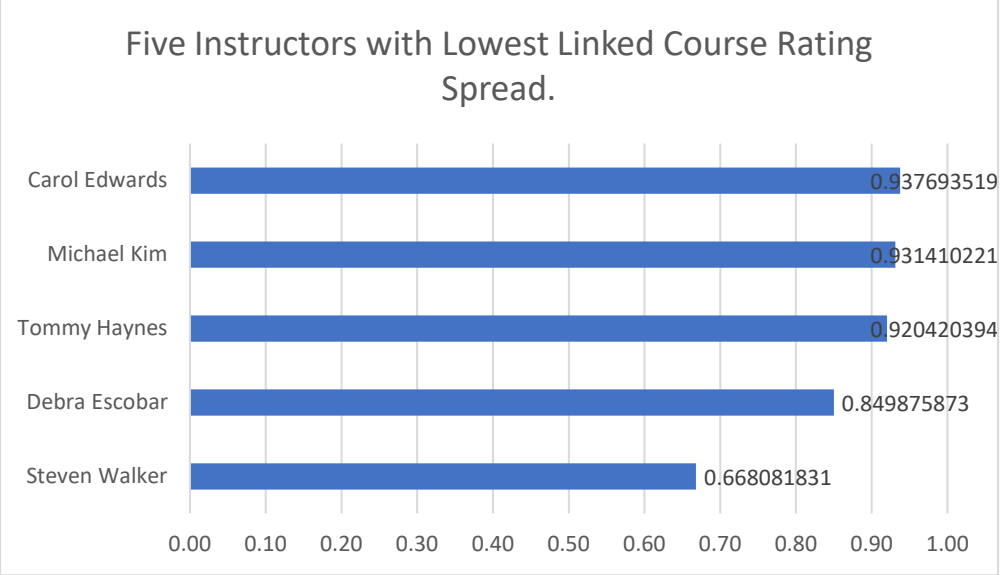
The distribution of instructor-course pairs across levels was similar for both genders. Advanced courses represented **41.24%** of female pairs and **38.95%** of male pairs. Beginner shares were **32.88%** and **35.47%**, respectively, while Intermediate shares were approximately **26%** for both.

These percentages describe recorded course relationships and do not measure teaching ability or establish differences in performance.

4.7 Linked Course Rating Consistency

Steven Walker had the smallest linked course rating standard deviation (**0.67**), with an average rating of **3.04 across 9 linked courses**. The platform's Rating Consistency Index, defined as the average of the 60 instructors' linked-course standard deviations, was **1.15**.

Lower standard deviation indicates less variation. However, consistently low ratings can also produce a small standard deviation. Therefore, spread should be interpreted alongside average ratings and course counts. This index describes course portfolios and is a proxy rather than a direct measure of instructor reliability.



Recommendations;

1. **Review lower-rated courses.** Prioritise courses in Machine Learning and Business, along with lower-rated Advanced courses. Examine learner feedback, content accuracy, lesson structure, and assessment difficulty before deciding what to change.

2. **Collect instructor-specific course feedback.** Record ratings for each instructor's delivery of a course, together with review counts and dates. This will help distinguish instructor performance from overall course quality.

3. **Provide training based on identified needs.** Review feedback and teaching practices for lower-rated instructors. Use confirmed gaps to design training and assess whether ratings improve afterward.

4. **Investigate enrollment concentration.** Examine why Yolanda Levine and Kimberly Miller account for 60.86% of recorded enrollments. Potential explanations to investigate include course availability, pricing, promotion, and visibility; the current analysis does not establish these causes.

5. **Evaluate quality using several measures.** Consider instructor ratings, linked course averages, rating variation, course counts, and learner feedback together. Experience and enrollment volume alone should not determine teaching quality.

6. **Monitor results regularly.** Use the deployed Streamlit dashboard to track performance by expertise, category, level, and rating range. Reassess results as new data becomes available.

Limitations

- Course ratings are shared across multiple instructors, so they cannot isolate an individual instructor's teaching impact.
- TeacherRating is a single recorded value per instructor. It cannot establish performance consistency over time.
- Learner review counts and individual feedback are unavailable, limiting assessment of rating reliability.
- Instructor-course pairs share courses and instructors, so the 887 pairs are not independent observations.
- Some expertise and experience groups contain few instructors or linked courses, limiting broad conclusions.
- Transactions represent recorded enrollments, not attendance, completion, learning outcomes, or repeat enrollment intentions.

- Correlations describe associations and do not establish causation. The analysis does not control for pricing, promotion, course availability, or other possible influences.
- The Experience Impact Score, Enrollment Influence Ratio, and Rating Consistency Index use definitions established for this project; they are not validated measures of causal impact or instructor reliability.

Conclusion

This study provides a descriptive framework for evaluating instructor ratings, linked course quality, and recorded enrollments on EduPro. Average teacher and course ratings were **3.13** and **3.10**, respectively. Experience showed a positive association with teacher ratings, while experience and teacher ratings showed almost no linear relationship with linked course ratings.

Course ratings varied across categories and levels, and enrollments were concentrated among two instructors. These findings support targeted course reviews and further investigation of demand patterns. However, shared course ratings limit conclusions about individual teaching effectiveness. Collecting instructor-specific feedback and monitoring results through the deployed Streamlit dashboard will strengthen future evaluation.

Live dashboard:
https://edupro-instructor-course-dashboard-s3e2gn8ekgxljhyguugslk.streamlit.app/

GitHub repository:
https://github.com/meghanagowda978-cell/edupro-instructor-course-dashboard